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Review – Prostate Cancer

Editorial by Rouvier Al-Monajjed, Peter Albers, Hein van Poppel, Ola Bratt, Monique J. Roobol on pp. 239–240 of this issue

Effect of Prostate Cancer Screening with Prostate-specific Antigen Testing on Long-term Prostate Cancer–specific Mortality: A Systematic Review and Meta-analysis of Randomized Controlled Trials

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Abstract

Context: Earlier evidence syntheses reached uncertain conclusions about the effect of prostate-specific antigen (PSA)-based screening on prostate cancer (PCa)-specific mortality (PCSM). Extended follow-up from major trials has provided an opportunity to re-evaluate the evidence.

Objective: To estimate the effect of PSA-based screening versus no screening or usual care on PCSM in individuals engaging in screening.

Evidence acquisition: We systematically searched PubMed, MEDLINE, EMBASE, and CENTRAL through October 2025 for randomized controlled trials (RCTs) comparing PSA-based screening with no screening or usual care among adults with prostates. The primary outcome was PCSM at the longest available follow-up. Secondary outcomes were PCSM beyond 12 yr and PCSM closest to 10 yr of follow-up. Risk of bias was assessed using the Risk Of Bias instrument for Use in SysTematic reviews-for RCTs (ROBUST-RCT). Incidence rate ratios were pooled using fixed-effect models. Certainty of evidence was evaluated using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach.

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Evidence synthesis: We included 5 RCTs analyzing 721,607 participants aged 45-80 yr of age with 11–23 yr of follow-up. PSA-based screening, with a high certainty in the evidence, demonstrated a reduced risk of PCSM at the longest follow-up ($p < 0.001$). Secondary analyses suggested greater relative PCSM reduction with longer follow-up. The overall effect was observed despite limitations including differing screening protocols and control-arm contamination, which likely underestimated benefit.
Conclusions: With 11–23 yr of follow-up, high-certainty evidence demonstrates that PSA-based PCa screening is associated with reduced PCSM among individuals who engage in screening—a finding that contrasts with earlier evidence syntheses. This finding does not in itself justify population-based screening or imply that every well-informed individual would elect to undergo screening. This PCSM benefit should be weighed against known harms and consider emerging approaches under study that may mitigate these harms.
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ADVANCING PRACTICE

What does this study add?
This study provides the most definitive synthesis to date of randomized controlled trial evidence on prostate-specific antigen (PSA)-based screening's effect on prostate cancer–specific mortality (PCSM) by incorporating long-term follow-up from all major trials. Unlike earlier syntheses of older trial data, this study demonstrates with high certainty that PSA-based screening reduces PCSM, with greater relative benefit observed at longer follow-up. The pooled absolute PCSM reduction should be interpreted cautiously in the context of heterogeneous trial designs that likely attenuated observed effects, weighed against the known harms of PSA-based screening, and considered alongside approaches now being developed to reduce these harms.

Clinical Relevance
This systematic review and meta-analysis of randomized controlled trials provides the most comprehensive synthesis to date of long-term evidence on PSA-based prostate cancer screening, demonstrating with high certainty that screening is associated with a reduction in cancer-specific mortality after 11–23 yr of follow-up. For uro-oncologists, these findings help clarify a long-standing controversy by confirming that the mortality benefit of PSA-baseds creening emerges and strengthens with extended follow-up, while also highlighting that earlier meta-analyses may have underestimated this effect. Clinically, the results support informed decision-making and guideline discussions around prostate cancer early detection, emphasizing that the modest mortality benefit must be balanced against screening-related harms and integrated within modern risk-adapted strategies. Associate Editor: Gianluca Giannarini, MD.

Patient Summary
These findings indicate that PSA-based screening lowers the risk of prostate-cancer death over 11 to 23 years of follow-up. These findings were judged to be of high certainty based on the updated evidence. The observed benefit should be cautiously understood in light of trial designs that likely made the effect appear smaller, weighed against the known harms of prostate cancer screening, and considered alongside novel approaches seeking to reduce these harms.

1. Introduction

The number of prostate cancer (PCa) deaths is expected to increase in the coming decades.[1] Although early detection through prostate-specific antigen (PSA) blood testing is possible, its benefits remain uncertain.[2,3] The major PCa-screening randomized controlled trials (RCTs) produced conflicting results.[4,5] Some showed no reduction in deaths from PCa, while there was also clear evidence of overdiagnosis and overtreatment associated with PSA-based screening.[6–9]

Over time, follow-up from these same RCTs has suggested that the disease-specific mortality benefit of screening takes many years to appear.[10–14] Despite these later findings, syntheses of the available RCT data at 10–15 yr of follow-up continued to conclude that PSA screening probably has little or no effect on PCa deaths.[7,15–17] These RCTs have now reported additional extended mortality results: the Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial (PLCO) in 2019, the Cluster Randomised Trial of PSA Testing for Prostate Cancer (CAP) in 2024, and the European Randomised Study of Screening for Prostate Cancer (ERSPC) in 2025.[12,13,18]

These new long-term follow-up data, as well as renewed interest in informing population-based PCa-screening

strategies and the early phases of new major RCTs in this space,[1,19–24] provide an impetus to revisit the evidence on PSA-based screening. We therefore conducted a systematic review and meta-analysis of RCTs to determine whether screening for PCa with PSA testing reduces PCa–specific mortality (PCSM) in individuals with a prostate engaging in screening.

2. Evidence acquisition

2.1. Review design

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement, the Cochrane Handbook for Systematic Reviews, and the Guidelines for Meta-analyses and Systematic Reviews in Urology. [25–27]

2.2. Protocol registration

We pre-registered our protocol with PROSPERO (CRD420251164229).

2.3. Patient involvement

An experienced research patient partner (EM), a PCa survivor, contributed to all stages of the study, from conceptualization to manuscript review.

2.4. Search strategy

We searched CENTRAL (Cochrane Central Register of Controlled Trials), Embase (via Ovid), MEDLINE (via Ovid), and PubMed from database inception to the most recent update, without language or date restrictions, on October 3, 2025. The search strategy, originally developed by a medical librarian and used in a previous review, was refined by the investigators.[7] A second medical librarian peer-reviewed the final strategy (Appendix 1). Before manuscript submission, we updated the search on October 31, 2025.

2.5. Inclusion criteria

We included published RCTs comparing PSA-based screening with usual care or no invitation to screening among individuals with a prostate and no prior diagnosis of PCa. We excluded RCTs reporting fewer than 8 yr of follow-up. No restrictions were applied by language or publication date.

2.6. Outcomes

The primary outcome was long-term PCSM, defined as death from PCa at the longest available follow-up among included RCTs.

Secondary outcomes included (1) PCSM beyond 12 yr of follow-up; and (2) PCSM between 8 and 12 yr of follow-up. For the former, we used the longest available estimate from RCTs reporting outcomes beyond 12 yr. For the latter, we used the estimate closest to 10 yr of follow-up. If two estimates were equidistant, we chose the latter one (eg, if the study reported 9-yr and 11-yr PCSM, we would retain the 11-yr estimate). The 8-yr threshold reflects understanding that PCSM benefits from screening require extended follow-up to manifest [7,14,16], while the 12-yr threshold corresponds to reporting intervals used in major PCa-screening trials and prior meta-analyses [7,10].

2.7. Study selection

After deduplication, we screened titles and abstracts using Covidence.[28] Two study team members independently assessed each record for eligibility, with disagreements resolved by a third reviewer. Reviewers also conducted full-text screening in duplicate, with conflicts adjudicated with a third reviewer. All retained reports were mapped to their trial.

2.8. Data extraction

Two study team members performed data extraction independently by using a standardized data abstraction form. We resolved discrepancies through consensus with a third reviewer. Extracted information included trial characteristics, participant characteristics, screening protocols, follow-up duration, and outcomes relevant to this review.

2.9. Large-language-model validation

Otto-SR (https://ottosr.com) is a large-language-model tool that supports systematic reviews after database search [29]. Blinded to human decisions, it independently screened search results and performed automated data extraction using the same standardized form. Its screening and extraction outputs were compared with the decisions of human reviewers. We will detail this process in a separate manuscript.

2.10. Risk-of-bias assessment

Two study team members independently assessed risk of bias using the Risk Of Bias Instrument for Use in SysTematic reviews-for randomised controlled trials (ROBUST-RCT) [30]. We did not use any of the optional ROBUST-RCT items. Since ROBUST-RCT does not address cluster-specific considerations, the Cochrane Risk of Bias 2 tool for cluster randomized trials (RoB 2 CRT) was used for the cluster RCT [31]. All discrepancies were resolved through consensus.

We summarized domain-level results graphically. After completing the domain-level assessments, we assigned each trial an overall risk-of-bias rating. This global judgment reflected our appraisal of how the identified biases were likely to influence PCSM [32]. We determined each study's overall risk of bias by consensus.

Blinding of participants and study investigators other than outcome adjudicators is unlikely to influence the outcome of PCSM. We, therefore, did not incorporate blinding of participants and study investigators in our overall judgement of bias.

2.11. Indirectness

Contamination of the control arm, that is when individuals in the control arm undergo PSA testing, is an important issue of PCa-screening RCTs [33]. Since contamination leads to more similar intervention and control groups, trials with

high contamination rate would report findings closer to no effect. Because contamination alters the applicability of the comparison to our question of interest (ie, the effect of PSA-based screening), we followed the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach which deals with contamination within the indirectness domain rather than the risk-of-bias domain [34].

2.12. Statistical analysis

We summarized dichotomous outcomes as incidence rate ratios (IRRs) with corresponding 95% confidence intervals (CIs). When RCTs reported person-years, we used the reported IRRs or calculated them directly.

We estimated the absolute risk reductions (ARRs) and number-needed-to-invite-to-screening (NNI) by applying pooled relative effects to a baseline risk. We did not prespecify a method for estimating this baseline risk of PCSM. We applied a conservative strategy. We pooled control-arm risks from trials with at least 12 yr of follow-up and lower contamination (ERSPC, CAP, and Stockholm). Because some degree of contamination almost certainly persisted in these control groups, this pooled value likely underestimates the true underlying baseline risk of these RCT populations. To support different decision contexts, we also presented ARR across a range of baseline risks from 0.01% to 5%.

We conducted all analyses using the intention-to-screen (ITS) principle. The Quebec trial reported per-protocol outcomes [35], so we extracted the available data to approximate ITS estimates.

We visually assessed statistical heterogeneity using forest plots. We also reported between-study variance (τ^2). Although we prespecified random-effects models in our protocol, the number of trials was too small to reliably estimate random-effects variance [27]. As such, we estimated pooled effects using fixed-effect models for our primary analysis. We used STATA MP18 (StataCorp, College Station, Texas).

2.13. Sensitivity analyses

We prespecified a sensitivity analysis restricted to RCTs at low risk of bias. This analysis assessed the stability of the effect when limited to the most methodologically robust evidence.

As a sensitivity analysis, we repeated the primary outcome analysis using a random-effects model to assess robustness to model choice.

2.14. Certainty of evidence assessment

We assessed the certainty of the evidence for each outcome using the GRADE approach [36,37]. Within GRADE, certainty-in-the-evidence ratings include high, moderate, low, and very low. We considered imprecision, inconsistency, risk of bias, publication bias, and indirectness [32,34,38,39]. The target of certainty of evidence rating was any effect of PSA-based screening on PCSM in individuals with a prostate engaging in screening [40].

All authors approved the final GRADE assessments by consensus. We presented Summary of Findings tables using the MAGICapp online platform (www.magicapp.org) [41].

3. Evidence synthesis

3.1. Study selection

The systematic search identified 6011 records. After deduplication, we screened 4109 records.

We excluded 4034 records based on titles and abstracts. We assessed 75 full-text articles for eligibility, of which 41 studies reporting on 5 unique RCTs met the eligibility criteria. We added one additional study published after our initial search. It reported the 23-yr follow-up of the ERSPC and was identified after the repeat search on October 31, 2025 [12]. Fig. 1 presents the PRISMA flow chart.

For the automated screening and data extraction with otto-SR, we found a high degree of concordance with the human reviewers. None of the discrepancies between otto-SR and human assessments affected the main findings. Full details of this validation will be reported separately.

Appendix 2 provides a list of representative full-text studies that were deemed potentially relevant but ultimately excluded.

3.2. Characteristics of included RCTs

The included studies corresponded to five unique RCTs (Table 1), with eight key publications aligned with the PCSM follow-up periods of interest (Supplementary Table 1) [4,12,13,18,35,42–44]. In total, these trials randomized 732 478 participants at baseline and analyzed 721 607 participants at their longest follow-up.

The included RCTs evaluated PSA-based screening in populations ranging from 45 to 80 yr of age. They differed in several key design features. For example, screening frequency varied across trials. The ERSPC and PLCO implemented repeated screening, whereas CAP and Stockholm evaluated the effect of a single invitation for one-time PSA testing. Contamination of the control arm varied substantially, with reported rates at different time points ranging from 7% to 90% [13,33,45–48]. Additional RCT characteristics are detailed in Table 1.

3.3. Risk-of-bias assessment

We judged that the ERSPC was the only RCT at a low risk of bias for the PCSM outcome. Table 2 presents the risk-of-bias assessment for each trial–outcome pair.

3.4. Baseline risk estimation

Studies with a lower rate of contamination and at least 12 yr of follow-up were ERSPC, CAP, and Stockholm. The baseline risks were 0.66% for CAP (15 yr of follow-up), 1.6% for ERSPC (23 yr of follow-up), and 3.1% for Stockholm (20 yr of follow-up). The pooled control group PCSM risk at their longest follow-up, weighted by cohort size, was 1.1%.

3.5. Effect of PSA-based screening on PCa-specific mortality at different follow-up lengths

3.5.1. Effect of PSA-based screening on PCa–specific mortality at the longest available follow-up

For our primary outcome, at the longest available follow-up from each included RCT, PSA-based screening was associ-

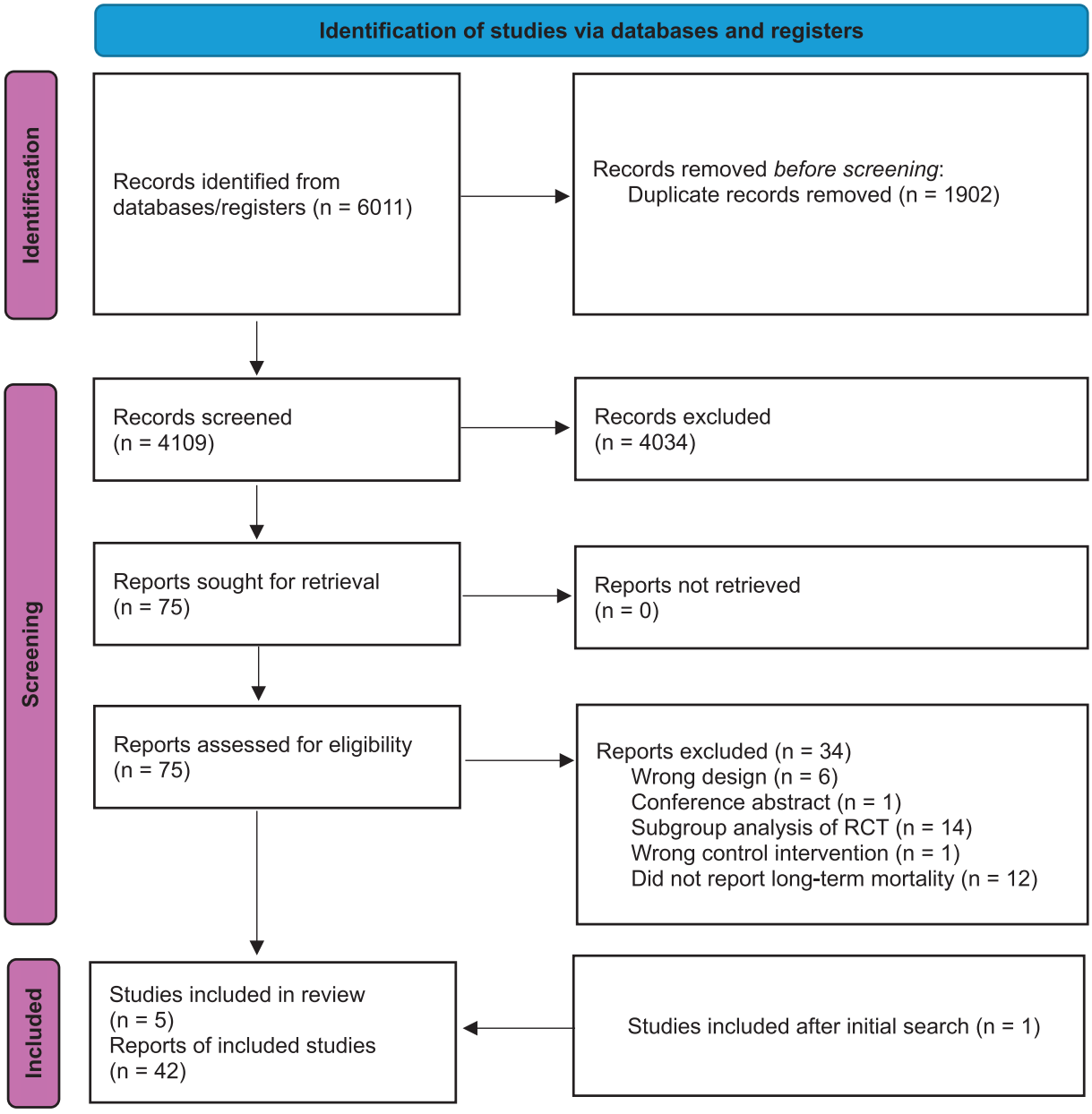


Fig. 1 – PRISMA flow chart.

ated with a 9% relative rate reduction in PCSM compared with no screening (IRR, 0.91; 95% CI, 0.87–0.96; $p < 0.001$; Fig. 2). This analysis included 721,607 participants across five trials, with follow-up durations ranging from 11 to 23 yr (median 17 yr). Between-study variance was negligible ($\tau 2 = 0$). Cochran's Q test was not statistically significant ($p = 0.4$). On visual assessment of the forest plot, effect estimate CIs across all trials included our pooled effect, with trials either showing a beneficial effect or a null effect.

Using the estimated baseline risk, PSA-based screening corresponded to an absolute reduction of 10 PCa deaths per 10 000 individuals screened with a follow-up of at least 8 yr (95% CI, 14 fewer to 4 fewer, high certainty in the evidence). This represents a NNI to prevent one PCa death of 1000 (95% CI, 700–2500). Supplementary Fig. 1 presents the ARR across baseline risks between 0.01% and 5% for the primary outcome.

3.5.2. Effect of PSA-based screening on PCa–specific mortality beyond 12 yr of follow-up and between 8 to 12 yr of follow-up

For the first secondary outcome, at the longest available follow-up beyond 12 yr (median 18.5 yr), PSA-based screening was associated with a 9% relative rate reduction in PCSM compared with no screening (IRR, 0.91; 95% CI, 0.86–0.96; $p < 0.001$; Fig. 3).

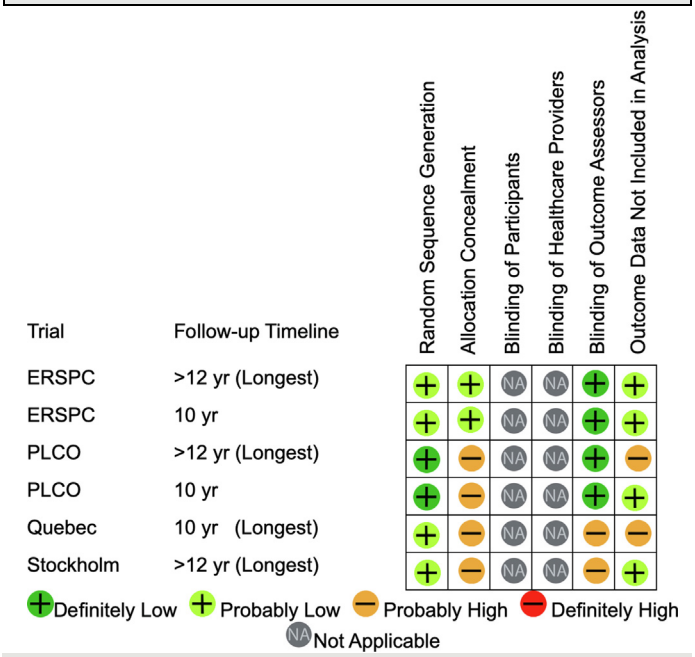
For the second secondary outcome, at the follow-up between 8 to 12 yr (median 10.5 yr), PSA-based screening was not associated with a statistically significant relative rate reduction in PCSM compared with no screening (IRR, 0.92; 95% CI, 0.84–1.00; $p = 0.050$; Fig. 4).

Table 1 – Characteristics of included trials.

Name (Enrolment yr)	Country	Design	Population	Number of individuals randomized (intervention/control)	Number of individuals analyzed at longest follow-up (intervention/control)	Screening and follow-up approach[a]	Biopsy Technique, compliance	Screening arm adherence	Longest reported mortality data[b]
CAP (2002–2009)	UK	Multicenter, Cluster RCT	Men aged 50–69 yr	197 938/221 644	189 326/219 395	One-time PSA PSA cut-off 3 ng/ml Subsequent trial-dictated diagnostic process	TRUS-guided systematic (10-core) Compliance: 85%	75,694 men (40%) men randomized to screening underwent PSA testing	15 yr
ERSPC (1993–2003)	Netherlands, Belgium, Sweden, Finland, Italy, Spain, Switzerland, France[c]	Multicenter, parallel RCT	Men aged 55–69 yr (core age group)	72 888/89 348	72 888/89 348	PSA+/-DRE every 2–4 yr PSA cut-off mostly 3 ng/ml Subsequent trial-dictated diagnostic process	TRUS-guided systematic (6–12 cores) Compliance: 89%	60,259 invited men (83%) received at least one screening	23 yr
PLCO (1993–2001)	United States	Multicenter, parallel RCT	Men aged 55–74 yr	38 343/38 350	38 340/38 343	PSA+/-DRE every year PSA cut-off 4 ng/ml Subsequent provider-dictated diagnostic process	Provider discretion Compliance: 41%[d]	32,591 invited men (85%) deemed compliant for PSA testing	17 yr
Quebec (1988)	Canada	Province-wide, parallel RCT	Men aged 45–80 yr	31 133/15 353	31 133/15 353	PSA +/- DRE every year PSA cut-off 3 ng/ml Subsequent trial-dictated diagnostic process	TRUS-guided systematic (6-core) with target if lesions identified on US Compliance: N/	7348 invited men (24%) were screened at the clinic	11 yr
Stockholm (1988)	Sweden	Regional, parallel RCT	Men aged 55–70 yr	2400/25 081	2400/25 081	One-time PSA, DRE, TRUS PSA greater than 7 ng/ml (palpation and TRUS) and 10 ng/ml (biopsy) Subsequent trial-dictated diagnostic process	TRUS-guided (2–4 cores ± FNAB), targeted if visible US or palpable lesions, otherwise systematic (4-core) Compliance: N/R	1779 invited men (74%) were screened	20 yr

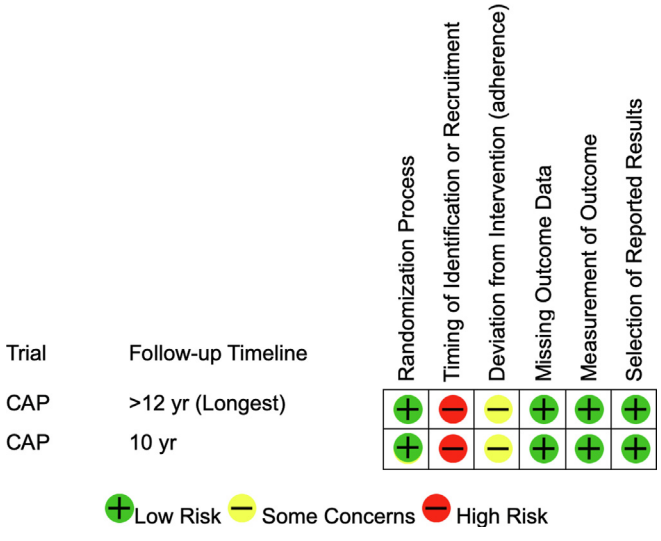
[a] The PSA cut-off refers to the value of the PSA that would prompt a biopsy and/or additional investigations.
[b] Refers to the longest follow-up for the full trial (ie, does not include subgroup analyses of men enrolled in earlier rounds with longer follow-up).
[c] Data from France were excluded from the ERSPC prostate cancer–specific mortality analyses due to failure to meet prespecified quality criteria and/or short follow-up duration.
[d] Unlike the other included trials, this information was obtained from an earlier report of the PLCO. The more recent reports did not report biopsy compliance.

Table 2a – Risk of bias in parallel RCT design (ROBUST-RCT tool)



We did not consider blinding of participants and investigators in the overall risk-of-bias rating as we did not consider this domain to influence the outcome of prostate cancer–specific mortality.

Table 2b – Risk of bias in cluster RCT design (ROB 2.0 for cluster RCT tool)



Detailed results for the secondary outcomes are presented in Appendix 3.

3.6. *Sensitivity analysis excluding studies at high risk of bias*

When we excluded trials at high risk of bias, only the ERSPC contributed data. The ERSPC's effect was consistent in direction and larger in magnitude than the pooled estimates across all studies (Supplementary Figs. 2–4).

3.7. *Sensitivity analysis using a random-effects model*

In the random-effects sensitivity analysis for the primary outcome, PSA-based screening at the longest available

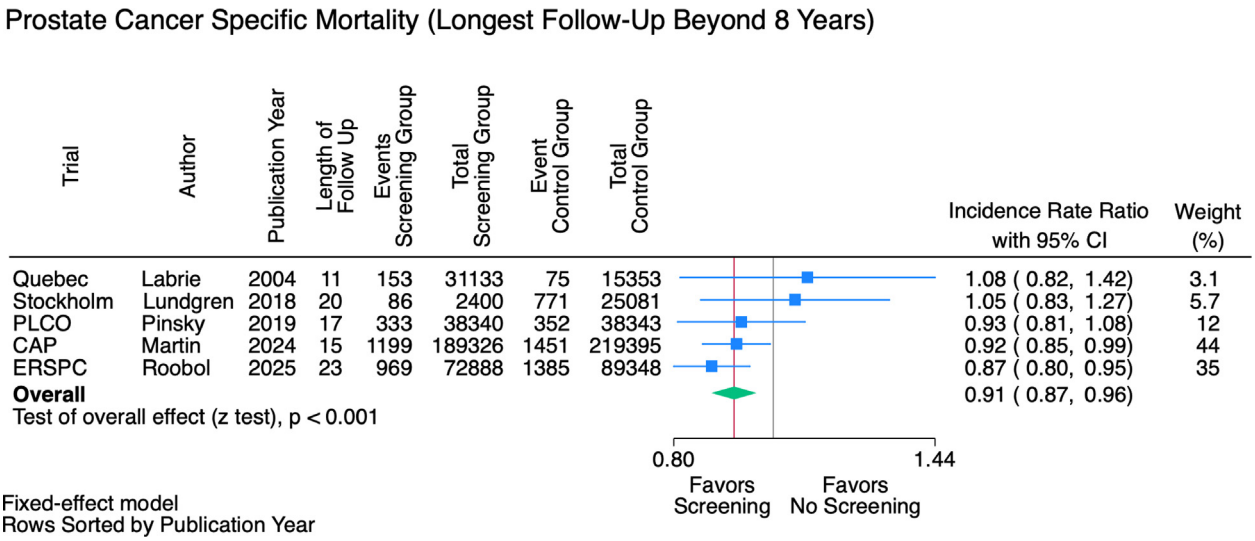


Fig. 2 – Forest plot of prostate cancer–specific-mortality rate ratios comparing PSA screening versus control at the longest available follow-up. Forest plot displaying the incidence rate ratios (IRRs) for prostate cancer–specific mortality comparing PSA screening with control groups. Each square represents an individual study, centered on its point estimate. Horizontal lines indicate 95% confidence intervals. The pooled IRR was derived using a fixed-effect model, and the diamond illustrates the overall summary estimate with its corresponding 95% confidence interval.

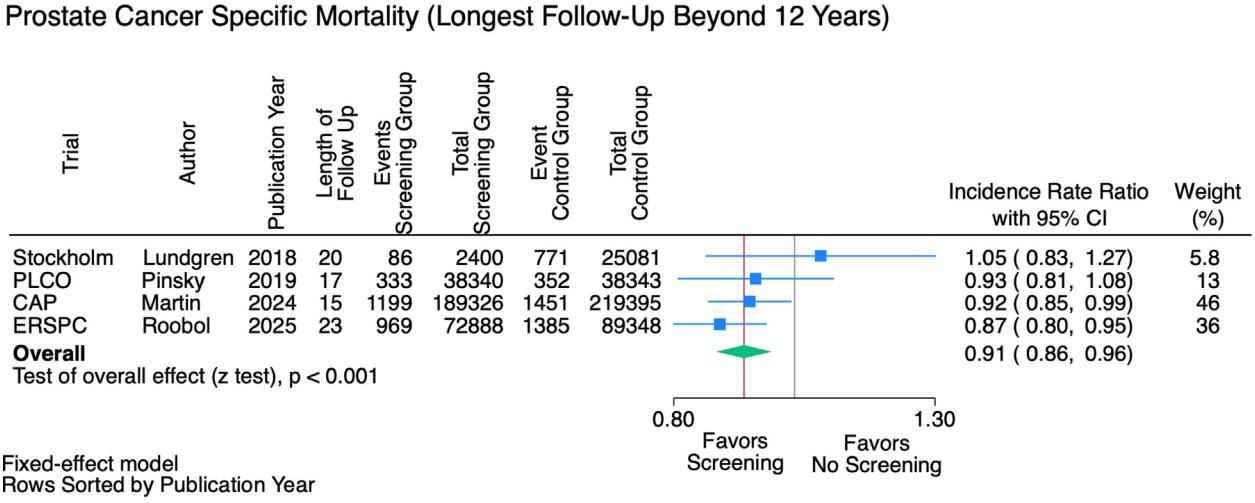


Fig. 3 – Forest plot of prostate cancer–specific-mortality rate ratios comparing PSA screening versus control beyond 12 yr of follow-up. Forest plot displaying the incidence rate ratios (IRRs) for prostate cancer–specific mortality comparing PSA screening with control groups. Each square represents an individual study, centered on its point estimate. Horizontal lines indicate 95% confidence intervals. The pooled IRR was derived using a fixed-effect model, and the diamond illustrates the overall summary estimate with its corresponding 95% confidence interval.

follow-up was associated with reduced PCSM compared with no screening (IRR, 0.92; 95% CI, 0.87–0.97; $p = 0.002$), consistent with the primary fixed-effect analysis.

3.8. *Grading of certainty of evidence*

The overall certainty of evidence for any benefit of PSA-based screening on PCSM was rated as high at the longest follow-up (Table 3). Our detailed GRADE rating is outlined in Table 4.

Certainty was also high for the outcome beyond 12 yr (Supplementary Table 2). In contrast, the 10-yr outcome was rated very low (Supplementary Table 2). Our detailed GRADE rating is outlined in Appendix 4.

4. Conclusions

In this systematic review and meta-analysis of PCa-screening RCTs, we found high-certainty evidence that PCa screening reduces long-term PCSM. The benefit became clearer with extended follow-up. Our rating of the evidence for our primary outcome is supported by an overall effect that excludes the null and by a plausible bias structure that would underestimate rather than exaggerate the true effect. Although control-arm contamination, nonadherence, and low-intensity interventions likely attenuated the observed effect, the disease-specific mortality benefit persisted. This finding does not in itself justify population-based screening

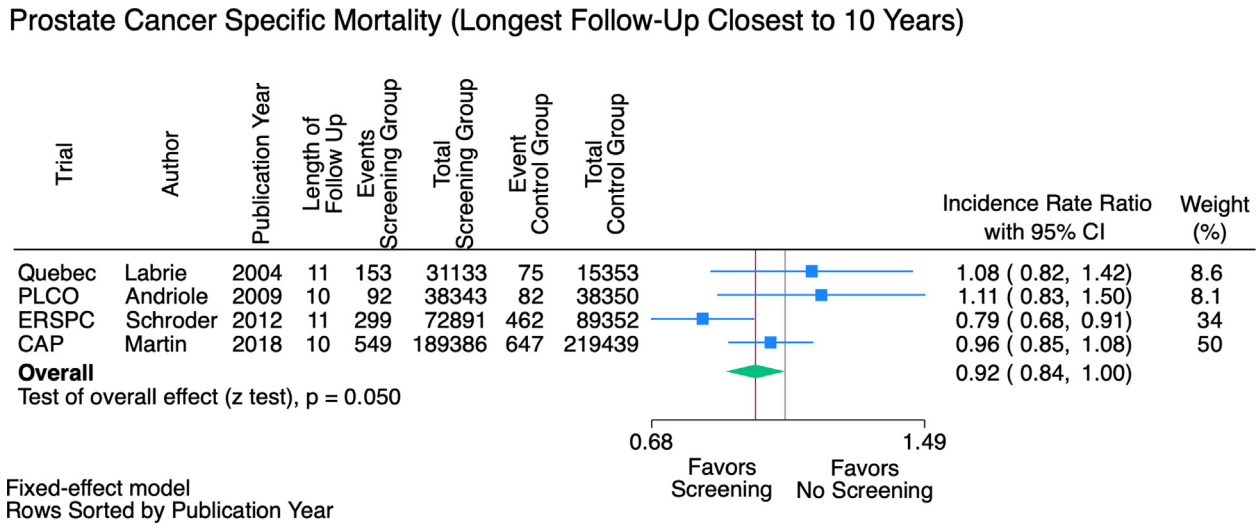


Fig. 4 – Forest plot of prostate cancer–specific-mortality rate ratios comparing PSA screening versus control at 8-12 yr of follow-up. Forest plot displaying the incidence rate ratios (IRRs) for prostate cancer–specific mortality comparing PSA screening with control groups. Each square represents an individual study, centered on its point estimate. Horizontal lines indicate 95% confidence intervals. The pooled IRR was derived using a fixed-effect model, and the diamond illustrates the overall summary estimate with its corresponding 95% confidence interval.

Table 3 – Summary of findings for the primary outcome

Outcome Timeframe	**Study results and measurements**	**Absolute effect estimates**		**Certainty of the evidence** (Quality of evidence)	**Summary**
		no screening	PSA screening		
Prostate cancer–specific mortality (beyond 8 yr of follow-up)	Rate ratio, 0.91 (95% CI, 0.87–0.96) Based on data from 724 007 participants in 5 studies Follow-up longest beyond 8 yr	**108** per 10.000	**98** per 10.000	**High** We did not rate down the certainty in the evidence for any of the domains.	PSA-based screening decreases prostate cancer–specific mortality (beyond 8 yr of follow-up)
		Difference: **10 fewer per 10.000** (95% CI, 14 fewer - 4 fewer)			

Population: Adult men.
Intervention: PSA-based screening.
Comparator: No invitation to screening.

Table 4 – Detailed reasoning of the rating of certainty of the evidence for the primary outcome

GRADE domain	**Rating**	**Rationale**
Starting certainty	High	The evidence base consisted solely of RCTs.
Indirectness	Not rated down	All trials had contamination of the control arm, which would be expected to attenuate rather than exaggerate any true effect of screening. Despite this, a prostate cancer–specific mortality benefit was still evident. Thus, the true benefit of screening (unaffected by control group contamination) in individuals engaging in screening would be similar or larger than our findings. Otherwise, the PICO questions from the primary trials aligned well with our review question.
Risk of bias	Not rated down	Four trials were deemed at a high risk of bias. The only trial with a low risk of bias yielded a more pronounced effect than the pooled estimate (Supplementary Fig. 2). This sensitivity analysis demonstrated the robustness of our findings. We therefore judged that the risk of bias was unlikely to account for the reduction in prostate cancer–specific mortality. When considered alongside indirectness, we concluded that the underestimation of the pooled effect due to contamination of the control arm outweighed any potential overestimation arising from the risk of bias domain.
Inconsistency	Not rated down	On visual assessment of the forest plot, effect estimate confidence intervals across all trials included our pooled effect, with trials either showing a beneficial effect or a null effect. The statistical measure of heterogeneity did not demonstrate considerable heterogeneity for our primary outcome.
Imprecision	Not rated down	The pooled estimate was precise given our target of rating was any versus no effect. Confidence intervals did not include a null effect of screening on prostate cancer–specific mortality. The total number of participants was large.
Publication bias	Not rated down	The evidence came from large, preregistered screening trials with complete reporting of mortality outcomes, and we found no signal of unpublished or selectively reported data. The low number of included RCTs precluded a funnel plot evaluation.
Overall certainty	High	In addition to the empiric steps listed above, the final certainty of evidence rating was reviewed by all study members and approved by consensus.

or imply that every well-informed individual would elect to undergo screening. This PCSM benefit should be weighed against the known harms of conventional PSA-based screening and consider emerging approaches under study that may mitigate these harms.

Our pooled estimate should be interpreted cautiously. The included trials varied substantially in their underlying interventions and designs, including the degree of control-arm contamination, screening intensity and frequency, biopsy thresholds, and contemporaneous treatment paradigms, among other factors. As such, the pooled estimate does not reflect the effect of a single, well-defined screening strategy, but rather the average effect observed across heterogeneous screening contexts. Despite these limitations which likely underestimate the pooled effect, the overall effect was statistically significant. Accordingly, we use the pooled estimate primarily to inform our assessment of the overall certainty and direction of the evidence supporting a reduction in PCSM associated with PSA-based screening, rather than to emphasize the precise magnitude of the summary effect.

Our finding of a PCSM benefit contrasts with previous systematic reviews and meta-analyses. The review by Ilic et al reported that screening may have no effect on PCSM [7]. It is important to mention that they emphasized the ERSPC as the trial at lower risk of bias and concluded that, at 13 yr of follow-up, it provided moderate-certainty evidence of a reduction in PCSM. Our updated finding was largely driven by the CAP trial's 15-yr results, which carried the greatest weight in our analysis and shifted away from the null compared to its 10-yr report [13,42]. The ERSPC remained consistently positive throughout its reports [5,10,12,44].

For the same reasons outlined previously, our NNI should be interpreted cautiously and likely represents an underestimate [12,49]. At 23 yr of follow-up, the ERSPC, the trial at lowest risk of bias included in our review, reported an NNI of 456 to prevent one PCa death. Absolute measures such as the NNI are sensitive to the underlying risk, duration of follow-up, and screening patterns of the population to which they are applied [50]. However, NNI values are frequently invoked in screening debates to compare the apparent efficiency of different cancer screening strategies. It is, therefore, useful to contextualize any NNI for PCa screening alongside those for other cancers. For example, the Canadian Task Force on Preventive Health Care reports a number-needed-to-screen of 1333 to prevent one breast cancer death among women aged 50–59 yr screened over 7 yr using mammography [51]. In an updated analysis of three Swedish breast cancer screening trials with over 20 yr of follow-up cited by the United States Preventive Task Force, the NNI to prevent a breast cancer death was 1539 [52,53]. In comparison, the NNI to prevent one colorectal cancer death using colonoscopy or guaiac fecal occult blood testing is significantly lower, ranging between 100–500 beyond 10 yr of follow-up [54–56]. It is noteworthy that the NNI estimate has decreased over time in trials such as the ERSPC; however, these absolute measures remain dependent on trial-specific baseline risk estimation methods and background screening intensity [12]. Nonetheless, the absolute benefit of conventional PSA-based screening should be weighed against the known harms of overdiagnosis and overtreatment [57]. In 2012, ERSPC investigators estimated that the harms of conventional annual PSA screening would be the overdiagnosis and overtreatment of 45 cases and the loss of over 1000 life years free of PCa over the lifetime of 1000 men [49].

We intentionally did not pool the RCTs' per-protocol or adherence-adjusted estimates, as such analyses differ in design, adjustment methods, and underlying assumptions, making an accurate interpretation challenging. These estimates generally suggest a larger benefit [12,13]. For example, Tsodikov et al accounted for contamination and lead-time to reconcile the effects of screening in the PLCO and the ERSPC [58]. They estimated a 30% lower risk of PCa death with PSA-based screening. In a target trial emulation using United States Medicare data, researchers evaluated the per-protocol effect and found that annual PSA screening over 8 yr was consistent with a reduction of approximately 10–40 fewer PCa deaths per 10 000 men aged 67–74 yr screened [59]. Taken altogether, despite the limitations of each approach, there is convergent evidence supporting an effect of PSA-based screening on PCSM.

We highlight several key limitations. First, contamination of control groups was substantial in several trials, reaching up to 90% in the PLCO [33], and likely worsening over time in most studies as PSA screening became more common [60]. However, the effects of contamination would likely shift results toward the null. Second, adherence to assigned screening in intervention arms was incomplete, further attenuating the observed effect. Third, the CAP and Stockholm trials employed low-intensity screening strategies, limited to a single PSA test, which likely underestimated the benefit achievable with repeated screening [14,61]. Collectively, these three limitations suggest that the true effect for individuals engaging in modern PCa screening may be greater than that reflected in our pooled estimates. The directionality of these biases was considered when rating our certainty of evidence for an overall effect on PCSM. Fourth, we did not capture secular effects associated with adoption of new early detection strategies, such as the use of magnetic resonance imaging and other novel adjunct tests. Older trials, particularly the Quebec and Stockholm studies, included outdated practices and are therefore unlikely to be directly generalizable to current early detection pathways. Fifth, some of the included RCTs did not report participants' race or ethnicity, or predominantly included white participants from North America and Europe, limiting their generalizability to more diverse populations and other settings with differing baseline risks of aggressive PCa [62]. Sixth, we focused on deaths from PCa, rather than overall mortality, because RCTs large enough to detect a difference in all-cause death require infeasibly large sample sizes, exceeding those of any individual study and those of our pooled analyses of RCTs [7,63,64]. Most common cancer screening tests, including mammography, fecal occult blood testing, and lung computed tomography fail to demonstrate that they save lives by extending lifetime [65]. In the absence of an unlikely increase in deaths from other causes due to screening, a real

reduction in PCSM implies a reduction in overall mortality [66,67]. We did not examine stage at diagnosis or overdiagnosis and overtreatment, since these outcomes are tied to the screening and diagnostic strategies used in each era. The trials with extended PCSM follow-up captured in our review used conventional PSA-based protocols, while newer studies using risk-stratified approaches are the ones positioned to modify the outcomes of stage at diagnosis, overdiagnosis, and overtreatment. However, these RCTs have not yet, or do not intend to, report PCSM [23,24,68]. Despite these limitations, our study provides the most current synthesis of the evidence and incorporates the latest updates from both the 23-yr ERSPC report and the 15-yr CAP report [12,13].

Our findings have implications for practice, policy, and future research. While previous evidence syntheses questioned whether PSA-based screening can reduce deaths from PCa [7,15–17], our study suggests with high certainty that it does with extended follow-up. We do not interpret our pooled estimate as representing the effect of a single, uniform screening strategy. Optimizing the delivery of PCa screening requires balancing benefit against the harms of overdiagnosis and overtreatment [6,49]. Important questions remain about whether screening should be implemented, and, if implemented, how it should be delivered, to whom, how often, and under what conditions. For example, the optimal age at which to initiate and discontinue screening remains uncertain. The PROBASE trial is evaluating a risk adapted screening strategy with an earlier starting age of 45 yr compared with 50 yr [69]. Emerging evidence suggests that low PSA levels measured at younger ages may inform long-term risk stratification [70], consistent with prior observational data demonstrating the prognostic value of a midlife PSA [71]. Appropriate PSA thresholds, including how they should vary across populations and in conjunction with adjunct testing, also remain to be defined [72]. Translation of trial findings into real-world screening programs must additionally account for barriers to participation, which may be particularly pronounced among individuals of lower socioeconomic status [73]. Modern approaches under investigation may achieve a balance by decoupling screening from immediate biopsy and treatment, and by incorporating secondary tests such as magnetic resonance imaging, micro-ultrasonography, and novel biomarkers [11,68,74–77]. Contemporary biopsy techniques, such as the transperineal approach, and improved antimicrobial prophylaxis have also markedly lowered the risk of sepsis, a major concern in earlier screening eras [78,79]. These refinements have the potential to reduce overdiagnosis, unnecessary procedures, and complications [80]. Moreover, active surveillance has become the preferred management approach for low-risk PCa, avoiding treatment-related morbidity without compromising quality of life or oncologic outcomes [81,82]. The ongoing challenge for researchers, clinicians, and policymakers is to design and evaluate screening strategies that maximize benefit, minimize harm, and align with current diagnostic and therapeutic paradigms [2,19–22].

In conclusion, with 11–23 yr of follow-up, the totality of evidence now demonstrates with a high level of certainty that PSA-based PCa screening in individuals reduces PCSM compared with no screening—a finding that contrasts with earlier evidence syntheses that did not establish such a benefit. These findings should guide shared and informed decision-making, inform policy development, and refine the design of future trials. Still, our finding of a PCSM benefit, considered in isolation, does not justify population-wide screening, nor does it imply that every well-informed individual would choose to be screened. Importantly, this disease-specific mortality benefit must be considered alongside the harms of screening, namely overdiagnosis and overtreatment, which have evolved with changing practice. Novel risk-stratified screening approaches are being investigated in ongoing trials designed to preserve this PCSM benefit of PSA screening while mitigating the known harms.

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David-Dan Nguyen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing - Original draft, Writing - Review & Editing.
Borna Tadayon Najafabadi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing - Original draft, Writing - Review & Editing.
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Alexandre R. Zlotta: Methodology, Investigation, Writing - Review & Editing.
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Laura C. Rosella: Methodology, Supervision, Investigation, Writing - Review & Editing.
Philipp Dahm: Conceptualization, Methodology, Supervision, Investigation, Writing - Review & Editing.
Gordon Guyatt: Conceptualization, Methodology, Supervision, Investigation, Writing - Review & Editing.
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Supplementary material

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